

SNEHA GHOSH

📍 Kolkata, West Bengal, India ✉ snehanew2003@gmail.com ☎ (+91) 9832178973 📁 Personal Portfolio in Sneha Ghosh
🔗 SnehaGhoshNew2003

OVERVIEW

B.Tech graduate and highly motivated Data Science enthusiast with hands-on experience in data preprocessing, machine learning, and deploying end-to-end analytical solutions using Python, SQL, and Streamlit. Proficient in EDA, model building, and result interpretation, with strong knowledge of tools such as TensorFlow, Keras, Power BI, and Langchain. Skilled in transforming business problems into data-driven insights. Seeking an AI/ML internship or entry-level role to contribute to impactful product development, thrive in fast-paced environments, and grow continuously in the field of applied data science.

EDUCATION

College of Engineering and Management, Kolaghat Bachelor of Technology in Information Technology CGPA: 8.25	2021 – 2025
Kendriya Vidyalaya Ranaghat Class XII, Science Percentage: 87.4	2020 – 2021
Kendriya Vidyalaya Ranaghat Class X, General Percentage: 85.8	2018 – 2019

SKILLS

Technical Skills : Python, SQL, Statistics, Probability, Linear Algebra, Calculus
Technical Tools : Power BI, Excel, Numpy, Pandas, Matplotlib, Tensorflow, Keras, Langchain, LangGraph, FastAPI, Docker, Vector Database, LLM, RAG, CrewAI, MCP, n8n
Soft Skills : Adaptability, Growth Mindset, Open to feedbacks, Communication Skills

CERTIFICATIONS

Udemy: The Data Science Course – Complete Data Science Bootcamp (2025) Key learnings:
◦ Data preprocessing, feature engineering, and statistical analysis. ◦ Implemented regression, classification, and clustering models using Scikit-Learn. ◦ Applied deep learning techniques using Tensorflow and Keras. ◦ Worked on real-world datasets for model building, evaluation, and tuning. ◦ Strengthened foundations in probability, linear algebra, and optimization.
TCS iON: IIT Kharagpur AI4ICPS Certificate Programme Highlights:
◦ Gained insights into industry-level AI applications and ML workflows. ◦ Studied key algorithms: SVM, KNN, Naïve Bayes, and clustering techniques. ◦ Explored deep learning, transformers, computer vision, and fine-tuning. ◦ Learned model evaluation, hyperparameter tuning, and real-world AI use cases. ◦ Understood ethical AI practices and deployment considerations.

EXPERIENCE

AI/ML Intern <i>Darpin Dynamics (Internship)</i>	December 2025 – April 2026
Voice AI – 24/7 Customer Calling System ◦ Built an end-to-end voice AI pipeline for automated 24/7 customer call handling using Voice AI Platform, enabling real-time conversational responses without human intervention. ◦ Integrated PostgreSQL as the backend data store for persisting call logs, customer records, and session states. ◦ Developed a companion analytics dashboard capturing call nuances, sentiment scores, multilingual transcripts, and customer feedback for business intelligence use.	
Multilingual Feedback AI System ◦ Built a feedback ingestion system that accepts customer feedback in any Indian language, performing automatic translation and transcription using speech and NLP models. ◦ Stored structured feedback data internally with sentiment tagging, enabling downstream analysis across regional language inputs. ◦ Designed an automated feedback categorization and tagging system to group user inputs into actionable categories, enabling faster decision-making and prioritization for business teams.	

Forensic Face Recognition & Restoration System

- Developed a forensic face recognition pipeline using **ResNet** for deep feature extraction and cosine similarity scoring, designed to identify individuals from degraded or post-mortem imagery with a preprocessing stage handling noise reduction and contrast normalization.
- Assisted in building a diffusion-model-based face restoration module using **GFPGAN** to reconstruct damaged or obscured facial images, enabling high-quality restoration as a pre-step to recognition.
- Evaluated pipeline robustness across varying levels of image degradation, validating recognition accuracy on challenging forensic-grade test cases.

Memory OS – Gmail & Calendar AI Assistant

- Designed a personal memory module that syncs Gmail and Google Calendar data using OAuth and stores indexed content via local persistence layer.
- Implemented a **RAG** (Retrieval-Augmented Generation) pipeline allowing users to ask natural language questions over their email and calendar history.
- Enabled AI-generated direct SQL queries from natural language prompts, allowing structured retrieval from synced email and event data stored in **PostgreSQL**.

Medical Bill PDF-to-JSON Extraction System

- Built a multi-stage intelligent pipeline to extract structured JSON from scanned and digital hospital bill PDFs using **PaddleOCR** for OCR and **pdfplumber** for digital text extraction.
- Integrated vision and language models via **Ollama** (**LLaVA**, **Qwen2.5**) to extract patient details, billing header, line items, and financial summaries with a custom legibility scoring module to auto-route between OCR and direct extraction paths.
- Exposed the pipeline as a **FastAPI** REST endpoint accepting PDF uploads and returning fully validated JSON; included deskewing, denoising, and category-aware batching for high accuracy on complex medical bill formats.

PROJECTS

TravelMind AI — LLM-Powered Travel Assistant

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- Architected a full-stack AI travel assistant using **FastAPI** backend with LLM API integration, supporting multi-turn contextual conversations for destination planning, route optimization, and real-time weather-aware recommendations.
- Designed a **RAG**-based question-answering module that retrieves location-specific context from a vector store, enabling accurate and grounded responses to complex travel queries.
- Built an interactive **Streamlit** frontend with dynamic session management, external API integrations (weather, geolocation), and real-time streaming responses for seamless UX.

Yoga Pose Detection & Real-Time Feedback System — Computer Vision

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- Built a two-stage computer vision pipeline: **YOLOv8** for real-time human keypoint and bounding box detection, followed by **VGG16** (transfer learning) for fine-grained pose classification across multiple yoga categories.
- Implemented a live corrective feedback engine that compares detected joint angles against reference pose vectors, overlaying visual correction cues on the video stream in real time.
- Improved model generalization through targeted data augmentation (flipping, rotation, brightness jitter), achieving robust performance across varied lighting conditions and body types.

Image Captioning System — Multimodal Deep Learning

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- Designed an encoder-decoder architecture using **Xception** CNN as a pretrained visual encoder and a custom **LSTM** decoder with attention-aware sequence generation, trained end-to-end on the **Flickr8k** dataset.
- Implemented beam search decoding (beam width = 3) as a replacement for greedy decoding, producing more fluent and semantically accurate captions with measurably improved BLEU scores.
- Deployed the full inference pipeline via **Streamlit**, supporting real-time image upload and caption generation with optimized preprocessing and tokenization for low-latency response.

Sentiment Analysis Engine — NLP / Sequence Modeling

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- Built a deep learning NLP pipeline using a stacked **LSTM** architecture with pre-trained word embeddings for fine-grained sentiment classification (positive / negative / neutral) on real-world customer review datasets.
- Engineered a robust text preprocessing pipeline with **NLTK**: tokenization, stopwords removal, stemming, and sequence padding — significantly improving model convergence and classification accuracy.
- Evaluated model performance across precision, recall, and F1-score metrics, with results demonstrating strong generalization on unseen review data from multiple product domains.

DASHBOARDS

Hospitality Domain: Created a Hospitality Domain Analysis using Power BI [\[See it here\]](#)

Sales Analysis: Created a Sales Insight using Power BI. [\[See it here\]](#)